BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021 • Paper Code: 92101

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 12

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. Which of the following is afghan scholar.

(A) Yaqub Kindi
(B) Ibnal Haitham
(C) Al beruni
(D) Abdul Qadeer.

Q2. The rate of change of displacement is called _____.

(A) Velocity
(B) Acceleration
(C) Force
(D) Speed

Q3. Which of the following forces can act as a centripetal force.

(A) Tension
(B) Friction
(C) Gravitational force
(D) All of these

Q4. The shortest distance between two couple forces is _____.

(A) Moment arm
(B) Couple arm
(C) Radius
(D) Double moment

Q5. The value of "g" at equator is _____.

(A) Same at poles
(B) Larger at poles
(C) Smaller at poles
(D) None of these

Q6. The product of mass and velocity is called _____.

(A) Momentum
(B) Force
(C) Torque
(D) None of these

Q7. Conventionally anti clock wise torque is taken is _____.

(A) Negative
(B) Positive
(C) Parallel
(D) Zero

Q8. The radius of earth is _____.

(A) 9.8 m
(B) 6.67×10^{-11}m
(C) 6×10^{24}m
(D) 6.4×10^{6}m

Q9. Work done will be zero when the angle between force and displacement is _____.

(A) 30°
(B) 45°
(C) 60°
(D) 90°

Q10. The unit of strain is _____.

(A) Kg m^{-3}
(B) Pa
(C) Nm^{-2}
(D) None

Q11. 310K in centigrade scale is _____.

(A) 37°C
(B) 310°C
(C) 63°C
(D) 273°C

Q12. The temperature at which a body is not radiating any heat is _____.

(A) 0c°
(B) 0°F
(C) 0k
(D) All of these

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 32

QII. Attempt any EIGHT Parts out of the following. Each Part carries equal marks.

(i) Distinguish between base and derived physical quantities.

(ii) Is it possible that displacement is zero but not the distance? Under what condition displacement will be equal to distance.

(iii) State and explain Newton third law of motion.

(iv) Why is the surface of conveyor belt made rough?

(v) Define like and unlike parallel forces with example.

(vi) Why is it more difficult to lean backwards? Explain.

(vii) Moon is attracted by the earth, why it does not fall on earth?

(viii) Define and explain Power.

(ix) Define and explain Pressure.

(x) A small needle sink in water and a huge ship travel easily in water without sinking?

(xi) Prove that $2as = vf^2 - vi^2$.

SECTION "C"

Maximum Marks: 21

Note: Attempt any THREE questions of the following. Each question carries equal Marks.

QIII.

(a) [4] Define scalar and vector quantities. Explain with example the graphical representation of vector quantities.

(b) [3] A bullet accelerate the length of the barrel of a gun 0.750m long with a magnitude of 5.35×10^3 m/s^2. With what speed does the bullet exist the barrel?

QIV.

(a) [4] Define equilibrium. Explain its types.

(b) [3] What are advantages and disadvantages of friction.

QV.

(a) [4] What is resolution of Forces? Explain with example how forces can be resolved into rectangular components.

(b) [3] At which altitude above Earth's surface would be gravitational acceleration be 4.9 m/s^2?

QVI.

(a) [4] Explain the law of Inertia with example.